
Specialist Physical AI Need Not Be Opaque

A Fully Tensor-Decomposable VLA Policy

Protocol-aligned control and exact weight analysis

Anonymous Author(s)

Affiliation

Address

email

Abstract

Tensor decomposition turns a large table of learned interactions into simpler low-rank factors that can be ranked and inspected. Conventional robot policies contain operations that block this direct weight-level view. We introduce χ -VLA, a rational vision-language-action model family with a 19.2M convolutional policy and a 20.1M ViT policy, both built only from bilinear, linear, and rational learned tensors rather than treated only as an opaque checkpoint. A runtime operator audit of the seed-0 20.1M ViT checkpoint and local numerical reconstructions verify its operator-level form. Across three independently trained 20.1M ViT policies, closed-loop LIBERO-Object success is $85.3\% \pm 4.6\%$ over 500 canonical trials per checkpoint. A fixed elementwise prediction-mean ensemble reaches $467/500 = 93.4\%$ at the cost of three forward passes. A separate convolutional instantiation passes an independent 4,608-case joint-attention certificate with worst relative error 6.36×10^{-16} . Exact-attention records naming those same ViT checkpoint artifacts numerically replay every attention module on one fixed cached input per checkpoint, with worst relative error 2.56×10^{-7} . In a seed-0 ViT checkpoint, exact weight-derived subspaces beat equal-rank random controls in all 36 primary block-head tests. The result is a capable and structurally inspectable specialist family with exact access to trained attention coefficients. It does not yet provide an exact end-to-end decomposition of the full policy.

1 Introduction

Modern robot policies can achieve high success while giving little indication of why an action was chosen. Their checkpoints contain millions of weights, but those weights do not directly show which image regions, words, and robot-state coordinates interact. Post-hoc probes can find correlated activations, yet a correlation is not automatically a mechanism or a safe editing target.

This is not a zero-shot policy. It is trained from scratch on one specialist suite. Its role in the workshop’s question is to establish an auditable specialist control against which future zero-shot and transfer claims can be evaluated.

A tensor is simply a multidimensional table. Tensor decomposition rewrites that table as a sum of low-rank factors, analogous to finding dominant directions in a matrix. If every learned block uses additions, multiplications, and ratios of polynomials, the full policy retains an explicit interaction table defined by its weights. Its decomposed factors then provide candidate control mechanisms that can be compared across layers and tested by intervention [3, 4]. This is what we mean by a fully

tensor-decomposable policy. It is an operator-closure claim, not a claim that we materialize one compact end-to-end coefficient tensor. Its degree and representation size grow rapidly with depth. For example, one coefficient can connect an image coordinate, an instruction coordinate, and an action coordinate. Factoring the coefficient table ranks high-energy cross-modal interactions directly from the checkpoint without first fitting an activation probe. Those factors are candidate structure, not automatically human-readable concepts or causal explanations.

The restriction is nontrivial in robotics. Softmax, ordinary activations, and square-root normalization break the algebraic form, while continuous actions require precise closed-loop outputs. The potential payoff is also concrete: a weight-derived mechanism could help diagnose whether a policy follows language, keys on a visual shortcut, or uses a fragile control feature.

We study the narrower but testable question:

Can a specialist VLA retain strong closed-loop capability when every deployed learned operation is polynomial or rational, while supporting exact weight analysis?

We test capability and structural analysis in two fully rational encoder instantiations. The same three ViT checkpoints establish closed-loop capability and support the exact attention audit. A separate convolutional instantiation tests whether structural certificates transfer across visual encoders.

Our contributions are:

1. We construct a complete VLA from tensor-compatible primitives. A rational normalization supplies input-specific scaling without leaving the rational function class.
2. We evaluate three fully rational 20.1M ViT policies on 1,500 canonical LIBERO-Object trials. They reach $85.3\% \pm 4.6\%$, while a fixed prediction-mean ensemble reaches $467/500 = 93.4\%$.
3. We mechanically audit the seed-0 20.1M ViT checkpoint and explicitly reconstruct deployed rational and bilinear branches from their weights.
4. We extend exact weight-derived decomposition through individual attention layers with a reduced-Gram calculation that avoids materializing the full high-order tensor.

This paper does not claim current state of the art. It also does not claim that comparable mechanisms are impossible to obtain from conventional policies. The empirical question is whether the algebraic policy makes exact weight structure directly accessible.

2 A rational tensor robot policy

2.1 Tensor-compatible primitives

Each primitive replaces a familiar transformer component while keeping explicit coefficients. Linear maps contribute first-order terms, while multiplying learned projections creates higher-order interactions that remain determined by the weights.

Homogeneous coordinate. We write $\bar{x} = [1; x]$. The constant coordinate allows a multiplicative layer to represent bias, linear, and quadratic terms in one contraction.

Bilinear feed-forward block. For $L, R \in \mathbb{R}^{r \times (d+1)}$ and $D \in \mathbb{R}^{d \times r}$,

$$\text{BFFN}(x) = D[(L\bar{x}) \odot (R\bar{x})], \quad y_o = \sum_{i,j} \left(\sum_{\rho} D_{o\rho} L_{\rho i} R_{\rho j} \right) \bar{x}_i \bar{x}_j. \quad (1)$$

The inner sum is a table indexed by one output and two input coordinates. The three matrices give its CP factorization, the tensor analogue of a low-rank matrix factorization.

74 **Softmax-free bilinear attention.** Each head forms two query-key systems and one value stream:

$$Y = C [M \odot (Q_1 K_1^\top) \odot (Q_2 K_2^\top) / s] V. \quad (2)$$

75 Here M is a fixed causal mask, C is fixed row scaling, and s is fixed. All learned query, key,
76 value, and output maps are linear. Equation 2 is therefore a polynomial in its input activations, with
77 interaction coefficients determined by the weights.

78 **Rational per-instance normalization.** RMS normalization contains an input-dependent square
79 root. Let $m(x) = d^{-1} \sum_{j=1}^d x_j^2 + \epsilon$, and let $s_0 > 0$ be the frozen running mean-square scale. We
80 deploy a fixed rational approximation $r(v) = P(v)/Q(v)$ to $v^{-1/2}$ over $v \in [0.1, 10]$:

$$v = \frac{m(x)}{s_0}, \quad \text{RNorm}(x) = \frac{x}{\sqrt{s_0}} r(v). \quad (3)$$

81 Here rational means a ratio of polynomials. Projective coordinates carry each activation as a
82 numerator-denominator pair (N, D) representing N/D . Updating this pair preserves input-specific
83 rescaling without leaving the rational tensor representation. Training and deployment use r itself. The
84 exactness claims below therefore concern faithful evaluation of the deployed rational computation,
85 not uniform agreement with RMSNorm. Appendix A.2 quantifies that distinction.

86 2.2 Architecture and training

87 Both encoder variants follow the high-level VLA sequence of visual encoding, language and state
88 embedding, joint processing, and action decoding [1]. They use a 64×64 agent-view image, an
89 eight-dimensional robot state, a 26-token task vocabulary, and eight action-query tokens. Both
90 share a width-384, eight-block, twelve-head joint backbone and a linear head that maps each action-
91 query state to seven continuous action dimensions. The 20.1M ViT uses four bilinear-attention vision
92 blocks and has 20,137,352 learned parameters. It supplies the closed-loop capability, runtime operator
93 census, exact-attention, and reduced-Gram results. The 19.2M convolutional variant instead uses three
94 stride-2 bilinear convolution stages that produce an 8×8 token grid. Three separate convolutional
95 files receive the joint-attention and rational-normalizer certificates reported in Appendix A.2.

96 Training uses 66,984 demonstration frames, 40,000 AdamW updates, batch size 256, cosine decay,
97 bfloat16 arithmetic, and an exponential moving average. Closed-loop evaluation uses a 180° camera
98 rotation to match the demonstration convention, canonical LIBERO initial states, and fail-loud checks
99 that verify every requested state was loaded.

100 2.3 Mechanical certificate on the seed-0 20.1M ViT checkpoint

101 Architecture definitions can differ from deployed code. We therefore audit the deployed seed-0 20.1M
102 ViT checkpoint at runtime and reconstruct representative operations from saved weights.

103 The normalization reconstruction has maximum absolute error 1.64×10^{-6} . A bilinear FFN re-
104 construction has median relative error 1.40×10^{-7} . These residuals are bounded by the deployed
105 module’s float32 statistic. The audit certifies operator membership and local reconstruction, as
106 summarized in Appendix Figure 3. It does not materialize one compact polynomial for the full
107 eight-layer transformer.

108 3 Protocol-aligned closed-loop capability

109 3.1 Evaluation protocol

110 We evaluate all ten LIBERO-Object tasks with the following fixed protocol:

- 111 • 50 canonical trials per task
- 112 • a 280-step cap
- 113 • three independently trained ViT checkpoints
- 114 • 500 rollouts per checkpoint and 1,500 rollouts in total

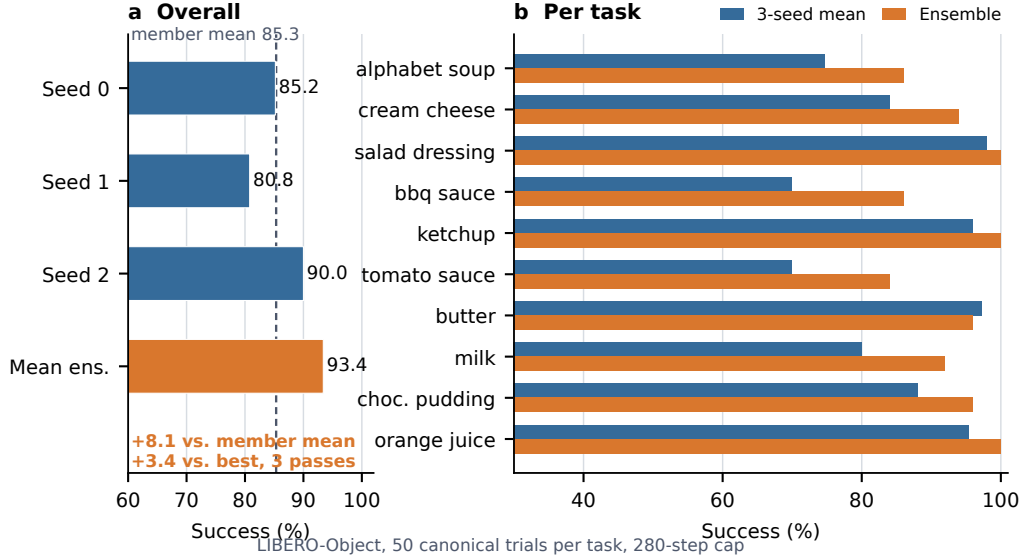


Figure 1: **Verified ViT capability.** Left: three independently trained checkpoints and their fixed elementwise prediction-mean ensemble. Right: per-task three-checkpoint mean and ensemble success. The ensemble uses three forward passes.

3.2 Results

Method	Parameters	Inference	Success
χ -VLA, rational ViT	20.1M	1 pass	85.3% $\pm$ 4.6%
χ -VLA, prediction-mean ensemble	$3 \times 20.1M$	3 passes	93.4%

Table 1: **Closed-loop LIBERO-Object success.** Each ViT checkpoint receives 500 canonical trials. The single-model row is the mean $\pm$ sample standard deviation across three checkpoints. The ensemble is one fixed composite evaluated on the same 500 states.

The three checkpoints reach $426/500 = 85.2\%$, $404/500 = 80.8\%$, and $450/500 = 90.0\%$. Elementwise averaging of their predicted action chunks reaches $467/500 = 93.4\%$. This is 8.1 points above the member mean and 3.4 points above the strongest member. The ensemble requires three complete forward passes and is not one 20.1M-parameter policy. Every episode-level record, task index, and protocol field is retained. The artifact manifest binds each immutable result file and recorded checkpoint basename to a checkpoint hash verified separately. Member and ensemble shards span different recorded GPU families and predate explicit matrix-precision fields, so the arithmetic gaps are descriptive rather than a matched-hardware ensemble ablation.

3.3 What this result does and does not establish

The result shows that the rational architecture supports high closed-loop specialist capability. It is an existence claim for a compact tensor-decomposable policy, not a claim of broad robotic generalization or state of the art.

4 Exact weight-derived structure through attention

4.1 Exact layerwise construction

Equation 2 contains two query-key products and one value stream. Expanding the linear maps gives a coefficient table whose entries measure how combinations of query, key, and value coordinates

132 contribute to an output. No examples or fitted probe are needed to construct it. On a tiny four-
 133 token, width-four, one-head layer, direct evaluation and an independent explicit polynomial agree to
 134 2.00×10^{-15} . The internal core identity agrees to 1.24×10^{-14} .

135 The deployed attention also normalizes each query and key with Equation 3. Each normalization
 136 multiplies all channels of one token by the same scalar rational factor. Those factors can be pulled
 137 outside the bilinear dot products. The polynomial attention core is unchanged, while explicit
 138 numerator and denominator factors carry the normalization. The rational extension agrees to $3.61 \times$
 139 10^{-16} . On one fixed cached input per checkpoint, we numerically replay all four vision and eight
 140 joint attention modules in each of three records that name the same ViT checkpoint artifacts as the
 141 capability results. This covers 36 module-input cases and 384 architectural heads. The worst relative
 142 ℓ_2 error is 2.56×10^{-7} and the worst absolute difference is 2.23×10^{-5} . This residual comes from
 143 the deployed implementation’s intentional float32 statistic, while the reconstruction itself uses the
 144 learned coefficients. The algebraic identity is input-general. The fixed inputs provide a numerical
 145 replay audit.

146 We separately certify three convolutional checkpoints from the same rational architecture family.
 147 Their vision stacks contain no attention, so we independently rebuild all eight joint modules and all
 148 twelve heads per module from the saved weights. Sixteen deterministically selected cache inputs
 149 per checkpoint yield 384 module cases and 4,608 head-input cases. Every output is finite and the
 150 worst relative ℓ_2 error is 6.36×10^{-16} , far below the frozen 10^{-6} gate. This same-checkpoint
 151 certificate covers the joint-attention equation at each normalized layer input. It does not rebuild the
 152 convolutional stem, feed-forward blocks, residual composition, final normalization, or action head as
 153 one end-to-end contraction.

154 Naively materializing the real width-384 coefficient tensor would require approximately 3.2×10^{15}
 155 values. We instead form an $O(d^2)$ reduced Gram. Like a covariance matrix, its leading eigenvectors
 156 rank important directions without constructing the full table. It matches brute-force materialization to
 157 2.13×10^{-14} at toy scale.

158 4.2 Trained policy result

159 The reconstruction certificate above spans three ViT checkpoints. The following subspace-ranking
 160 experiment is a descriptive analysis of the seed-0 ViT checkpoint. We apply the exact weight-derived
 161 calculation to all twelve heads in early, middle-late, and final backbone blocks 0, 6, and 7. Random
 162 projector banks are sampled once, then fixed and reused for all examples. The weights choose the
 163 subspace, while cached activations are used only afterward to score how well it preserves the layer’s
 164 computation.

165 At ranks 8, 32, and 128, respectively, 80.6%, 91.7%, and 100% of the 36 block-head pairs favor the
 166 exact subspace. Their median ratios are 1.028, 1.171, and 2.441. At rank 128, the mean ratio is 3.033
 167 and the strongest head reaches 10.135. The rank-128 block means are 3.08, 3.45, and 2.56 for blocks
 168 0, 6, and 7. The 36 block-head pairs are architectural units within one checkpoint, not independent
 169 model-level replicates.

170 This extends exact weight-only construction through individual attention layers. It is not an exact
 171 end-to-end decomposition of the complete policy. Composition across all eight blocks still faces
 172 rapid degree and representation growth.

173 5 Related work and limitations

174 **Related work.** OpenVLA and OpenVLA-OFT establish pretrained VLA and continuous-control
 175 baselines [1, 2]. Bilinear MLPs, χ -nets, and bilinear IMPALA policies show that multiplicative
 176 weights can expose low-rank structure and support subsequent controls [3–5]. Recent work also
 177 steers pretrained VLAs through activation directions [6]. Our distinction is the exact weight-derived
 178 construction in a policy designed as a rational tensor program. Tensor and polynomial networks
 179 provide the broader multilinear foundation [12–14]. VLA robustness benchmarks show that standard
 180 LIBERO success can coexist with weak language use [9–11].

181 **Limitations and conclusion.** The benchmark is one specialist suite, so it does not establish broad
 182 robotic generalization. Exact reconstruction is certified layer by layer and does not yet produce one

183 compact symbolic contraction of the full policy. The ViT records connect capability, exact attention,
184 and the reduced-Gram screen through common checkpoint basenames. Separate convolutional files
185 replicate the joint-attention equation certificate across another visual encoder. Evaluation uses fixed
186 task instructions and does not establish robust instruction grounding. Within that scope, rational
187 restrictions support strong LIBERO-Object control and exact trained-weight certificates. We do not
188 claim a direct coefficient-edit interface.

189 **Reproducibility.** The anonymous artifact releases raw structural-certificate JSONs and all 2,000
190 episode rows behind the ViT member and ensemble results. Immutable SHA-256 manifests and
191 standard-library verifiers independently recompute the capability totals and the convolutional-attention
192 and RationalNorm certificate decisions. Historical Modal cache-backed analyses read dataset-native
193 task metadata. The replacement Athena pipeline pins that metadata and maps dataset identifiers to
194 official identifiers before defining task splits. A source-level audit compared all 271,996 cached
195 frames across Object, Spatial, Goal, and LIBERO-10 with their pinned LeRobot revisions. Task
196 identifiers, language joins, states, actions, and resized images matched exactly for every frame.
197 Canonical-state checks fail loudly when requested states are unavailable.

References

- [1] M. J. Kim et al. *OpenVLA: An Open-Source Vision-Language-Action Model*. CoRL, 2024.
- [2] M. J. Kim, C. Finn, and P. Liang. *Fine-Tuning Vision-Language-Action Models: Optimizing Speed and Success*. 2025.
- [3] A. Rigg, T. Doods, M. T. Pearce, J. Oramas, and L. Sharkey. *Bilinear MLPs Enable Weight-Based Mechanistic Interpretability*. ICLR, 2025.
- [4] T. Doods, W. Gauderis, G. A. Wiggins, and J. Oramas. *Compositionality Unlocks Deep Interpretable Models*. 2025.
- [5] F. Oozeer, Y. Eriskin, A. Rigg, J. Oramas, and L. Sharkey. *Bilinear Convolution Decomposition for Causal RL Interpretability*. 2024.
- [6] B. Häon, K. Stocking, I. Chuang, and C. Tomlin. *Mechanistic Interpretability for Steering Vision-Language-Action Models*. 2025.
- [7] S. Marks et al. *Sparse Feature Circuits: Discovering and Editing Interpretable Causal Graphs in Language Models*. ICLR, 2025.
- [8] M. Kekić et al. *Learning Nonlinear Causal Reductions to Explain Reinforcement Learning Policies*. ICLR, 2026.
- [9] X. Zhou et al. *LIBERO-PRO: Towards Robust and Fair Evaluation of Vision-Language-Action Models Beyond Memorization*. 2025.
- [10] Y. Fang et al. *When Vision Overrides Language: Evaluating and Mitigating Counterfactual Failures in VLAs*. 2026.
- [11] C. Kim et al. *LIBERO-Para: A Diagnostic Benchmark and Metrics for Paraphrase Robustness in VLA Models*. 2026.
- [12] T. Kolda and B. Bader. *Tensor Decompositions and Applications*. SIAM Review, 2009.
- [13] E. Stoudenmire and D. Schwab. *Supervised Learning with Tensor Networks*. NeurIPS, 2016.
- [14] G. Chrysos et al. *Deep Polynomial Neural Networks*. CVPR and TPAMI, 2020 to 2021.

A Supporting results and decisions

A.1 Expanded attention screen

Within the same seed-0 ViT checkpoint, the corrected fixed-control attention analysis covers all eight blocks and twelve heads. At rank 128, 83/96 block-head pairs favor the exact weight-derived subspace. The median ratio is 1.94, the mean is 2.58, and blocks 2 and 3 provide useful depth controls. At ranks 8 and 32, only 54/96 and 60/96 pairs favor the exact subspace. The effect is therefore depth-dependent and strongest at the wider tested rank.

A.2 Numerical scope of rational conversion

The deployed degree- $[2/2]$ least-squares rational fit is exact as a rational computation even though it approximates RMS normalization. Its numerator has coefficients (5.32997, 7.53552, 0.327274) and the denominator has coefficients (1, 9.64827, 2.60664). Every denominator coefficient is positive, which certifies $Q(v) > 0$ for all $v \geq 0$. We instrument all 53 RationalNorm module instances, each invoked once per forward pass, on 4,096 deterministic inputs for each of three convolutional checkpoints. Across 544,542,720 activation rows, the minimum observed denominator is 1.0016 and no row is nonfinite. Comparing deployed float32 normalization with an otherwise identical pass that recomputes only RationalNorm scales in float64, the largest checkpoint-level primary metric, defined as the maximum of normalized- and physical-action NRMSE, is 2.414×10^{-6} against a frozen 10^{-3} gate.

241 The certificate also bounds what the operator approximates. Observed normalized mean-square ratios
 242 v span 1.66×10^{-4} to 17.17, outside the nominal fit interval $[0.1, 10]$. Across site-checkpoint pairs,
 243 the minimum coverages are 60.018% in that interval and 60.022% with rational-versus-exact-RMS
 244 scale error at most 3.4%. The frozen 99% secondary coverage gates therefore do not pass. The
 245 positive claim is therefore global absence of nonnegative denominator poles and cache-conditioned
 246 floating-point fidelity to the deployed rational operator, not uniform agreement with exact RMS
 247 scaling. Matched alternative-normalizer runtime overhead remains unmeasured.

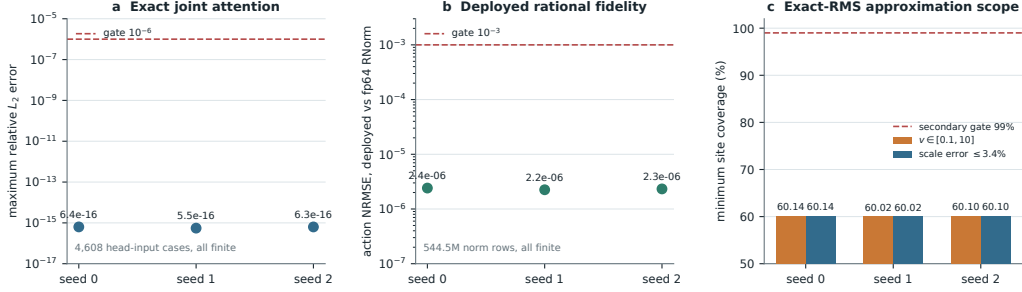


Figure 2: **Frozen structural certificates on three independent convolutional checkpoints.** Panels (a) and (b) pass their primary gates with large margins. Panel (a) reconstructs all 4,608 audited joint-attention head-input cases. Panel (b) compares deployed float32 normalization with an otherwise identical pass that recomputes only RationalNorm scales in float64. Panel (c) reports the separately frozen exact-RMS approximation coverage, which does not pass its 99% secondary gate.

Mechanical operator audit of the seed-0 20.1M ViT checkpoint

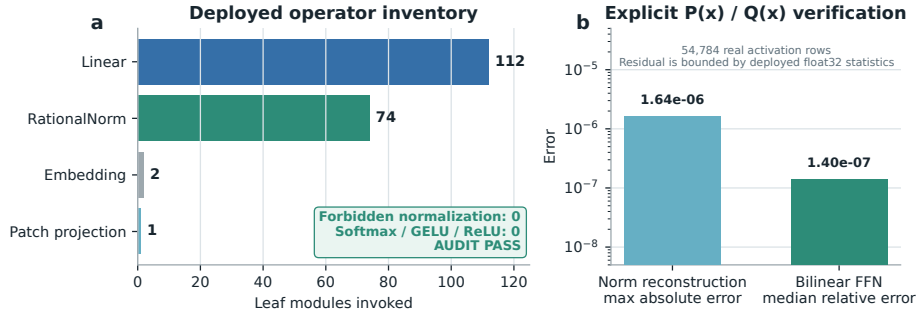


Figure 3: **Mechanical audit of the seed-0 20.1M ViT rational checkpoint.** The runtime path invokes 112 linear maps, 74 rational normalizers, two embeddings, and one linear patch projection. No softmax, standard activation, or ordinary normalization is present. Reconstruction errors are measured over 54,784 real activation rows.